zine

source

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Zine is a human-computer interaction language for philosophy. It is interesting in that its syntax is natural people language; in fact, most human languages can be used. Nevertheless, it is precise.

Figure 1. Zines from the seventies.[1]

Zine can also mean Zine Is Not Emacs, which I wrote this in.

Zine is also a very good what-you-see-is-what-you-get editor for Typst[2] technical writing.

With the widespread usage of LLMs the bullshit machine is running at full steam, and nuance is swept under probabalism.

If we contrapose the continental traditions with the analytical tradition, broadly taken to include mathematics all the natural sciences. In the analytical tradition, the rejection of accepting all statements as true is so strong that there is .

Even a mediocre engineer can contribute, while philosophy still takes enough discipline of the mind that prevention of falsehood is accomplished in the field through massive exclusion.

In physics, the tenth grade student can understand the Rutherford model of the atom and accept it as a fact upon which to build without

Science requires fixing a model first

Haskell is the glue, so that you can think about the parts in isolation. Category theory style Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

This resolves a nominal into a specific noun

```
resolve :: Nominal → Noun
```

These are all the thirty-seven relations in the Universal Dependencies grammar[3].

```
data Relation = Nsubj
              | Obj
              | Iobj
              deriving (Eq, Ord, Enum)

newtype Dependency source target = Dependency (source → ([Relation], target))
```

Because Dependency is generic on any source or target, coreference can be handled separately on a PoS layer I bet

```
–instance Category Dependency where
–  id = Dependency $ \source → ([], source)
–  (Dependency a) . (Dependency b) = Dependency $ \source →
–    let (relations, intermediate) = b source
–        (relations', target) = a intermediate
–    in (relations ++ relations', target)
```

Dependency parsing happens sentence-by-sentence

```
type Syntax = [Sentence]
```

UDPipe (maybe see https://github.com/fpco/inline-c), lazily

```
dependencyParse :: Text → Syntax
```

Functor from the tree-category of Dependency to a meaningful category. See below Section 2.2 – use Prolog

```
understand :: Syntax → Maybe Semantics

explain :: Semantics → Syntax
```

Maxima here, which needs a ECL FFI or general Lisp IPC

```
simplify :: Math → Math – Maxima
```

Yi, The keybindings part

```
interpret :: Input → Command
```

Yi, stuff from EditorM for example

```
execute :: Command → State → State
```

Custom, using typst-layout, which needs rust FFI or some (??) IPC

```
view :: State → DisplayList
```

WebRender ofc, which needs rust FFI (see https://github.com/harpocrates/inline-rust) or general Rust IPC

webrender :: **DisplayList → GL**

typstLayout :: **Typst.Syntax.Markup → DisplayList**

createFrame :: **IO GLContext**

read :: **IO Input**

The main interaction loop

```
main :: IO ()
main = do
  frame ← createFrame
  let interactionLoop state = do
        display ((webrender . view) state) frame
        input ← read
        let state' = (execute . interpret) input state
        interactionLoop state'
  interactionLoop initialState
```

Profane definitions of types

```
newtype GL = GL { display :: GLContext → IO () }
```

Haskell default and glue:
- Prolog (NLP Topos FOL). Maybe this can be in native Haskell or scheme
- Lisp (CAS) Better interop with scheme
- Rust (rendering)
- C++ (text tagging, Firefox IPDL) Can be done with udpipe-rs

lol maybe idris instead of haskell -> scheme -> rust marwood or other scheme implementation for display list, udpipe-rs -> prolog with call/cc -> lisp direct interop with scheme lists but embedded via ecl2

if they're all separate, you could get a prolog repl, an ECL repl, and a GHCI repl

maybe prolog translates directly into lispy things

haskell -> prolog -> lisp

1. Philosophy and sheaf theory

For example, epistemology, at its most basic, can be considered

$$\text{Fact} \longrightarrow \text{Knowledge}$$

Need to think about the pushout of rooted words, which would connect subjects to objects via some actual relation. Because of the categorical structure of Dependency, only the dependency paths between two different tokens matters. Pushouts can be related to specific morphisms, and will probably be through things like verbs

If I was to use sheaf theory here, then I've broken the problem into:
- Local semantics
- gluing, including coreference probably
- Global semantics

Globally, I want to extract stuff like

```
newtype Epistemology = Epistemology (Set Fact → Knowledge)
```

What is true about both the local semantics and the global semantics? Both subcategories of Hask, likely, as they both need to be representable internally by the computer ultimately.

So what is Epistemology? It's a subcategory with only two objects and it goes unidirectionally. pushouts are what two epistemic systems have in common and pullbacks are the dual

what about hegel

$$\mathit{dialectic} :: (\mathbf{Thesis}, \mathbf{Thesis}) \to \mathbf{Maybe\ Thesis}$$

metaphysics: what is? epistemology: what is knowledge? ethics: what is just/right/needful?

probably can't constrain this too much – maybe it's literally just collections of morphisms between two Hask objects

$$\mathbf{type}\ \mathrm{Philosophy\ a\ b} = [\mathrm{a} \to \mathrm{b}]$$

ok now we might be getting somewhere, as I can also make morphisms from local semantics via pushouts through a tree root in the dependency parse. I don't necessarily have to ask from whence nouns came from (this is restricted to humans), just the relationships defined between them.

2. Category theory

2.1. Definitions

If you're already familiar with categories and their representation in Haskell, you can skip this section. (There is already Control.Category, but it is necessarily a *wide* category, where its objects are always Ob(**Hask**).)

Haskell categories

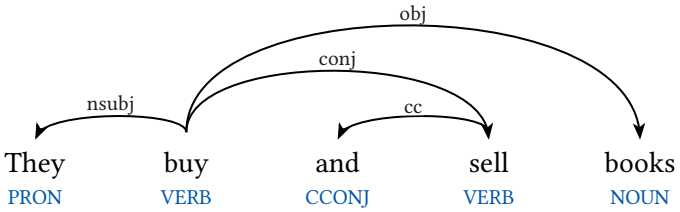
```
class Category (k :: Type → Type → Type) where
  type Object k (a :: Type) :: Constraint

  id :: Object k a ⇒ k a a

  (.) :: (Object k a, Object k b, Object k c) ⇒ k b c → k a b → k a c

instance Category (→) where
  type Object (→) a = ()
  id x = x
  (g . f) x = g (f x)
```

```
mor(A, A).
mor(A, C) :- mor(A, B), mor(B, C).
```

2.2. A Dependency Tree Topos

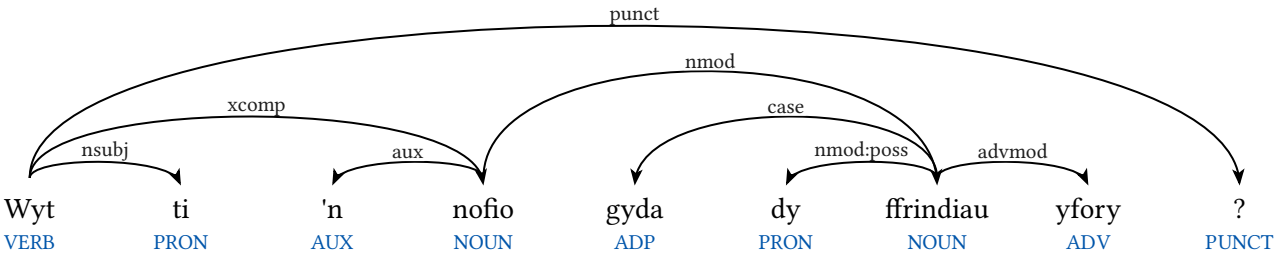
This is the core (Hegelian and category-theoretic) claim: **every noun is a verb-morphism between two other nouns**. For example, Nothing $\xrightarrow{\text{Knowledge}}$ Something.

Consider the dependency tree parse of the following sentence



In prolog

```
pron(word("They", 1)).
verb(word("buy", 2)).
cconj(word("and", 3)).
verb(word("sell", 4)).
noun(word("books", 5)).

dep(word("buy", 2), nsubj, word("They", 1)).
dep(word("buy", 2), conj, word("sell", 4)).
dep(word("buy", 2), obj, word("books", 5)).
dep(word("sell", 4), cc, word("and", 3)).

% Category definition
mor(word(String1, Index1), word(String2, Index2)) :- .
```



Let's define the type of dependency trees.

```
data DependencyTree a = DependencyTree {
  word :: a,
  children :: [(Relation, DependencyTree a)]
} deriving Functor
```

From this, you could construct the free category $\mathcal{D}$ with paths as morphisms. We get the following morphisms in **Cat**:

$$\mathcal{D} \xhookrightarrow{T} \mathcal{T}$$

To any dependency parse, we can associate a category, say $\mathcal{D}$, by considering the objects as words (and their indices) and the morphisms as dependencies. To be precise, $\mathrm{Ob}(\mathcal{D}) = \{(\text{They}, 1), (\text{buy}, 2), \ldots\}$

and $\mathrm{mor}(\mathcal{D}) = \{((\mathrm{buy}, 2), [nsubj], (\mathrm{They}, 1)), ...\}$. I'll use $\mathrm{buy}_2 \xrightarrow{\mathrm{nsubj}} \mathrm{They}_2$ as notation for $\{((\mathrm{buy}, 2), nsubj, (\mathrm{They}, 1))\} \in \mathrm{mor}(\mathcal{D})$ from here forward.

Topos of local truth constructed in just one sentence. First add necessary pushouts of morphisms through relation words to construct just noun phrases as objects in the category. Objects are nominals that are related to

The category of elementary topoi **Top** is a 2-subcategory of the 2-category **Cat**

Maybe– Terminal object: "this"/unrestricted PRON Binary products: different sentences, indicates that two independent clauses can be broken up Equalizer: CCONJ/union Exponential: linguistic recursion/which Omega: "itself" all nouns Sub object classifier: X–copula–>"itself"

Existential quantification would use "this", whereas the default is just universal. i.e. Red is bright vs this red is bright.

Now that I have a sheaf topos, I would like to be able to glue it together into a general understanding. The glue will inevitably have a nonzero cohomology in which case we will fail to synthesize a global understanding. But say we have a global understanding; then we can understand how different parts fit together by alternating between natural language representation and the categorical.

Once I have a local topos, I can attempt gluing using some kind of ACL2 combo or H-M unification, or both. I will need to figure this out later, and the exact system by which the local, overfit, understanding gets generalized to the global will determine the cohomology.

Maybe have a two-level sheaf attempting to glue sentences together in one argument, using some natural deduction and coreference (H-M unification there?) in the glue between sentences which are part of a topos. Then, can try a different kind of glue to see how two arguments fit together, and which way the functors go between them. Maybe the higher level isn't even a sheaf and just a topology defined on it.

3. Feeling the elephant

4. Diagrams

4.1. 3D

Appendix A. Referenced packages

The following imports were used in this Literate Haskell source file.

```
module Main(main, parseOnce, parseConllu, Doc) where

import Data.Kind (Constraint, Type)
import Prelude hiding (id, (.), read)
import qualified Language.C.Inline.Cpp as Cpp
```

Typst is used for parsing

```
import qualified Typst.Syntax

import Foreign.C.String (peekCString, withCString)
import Foreign.Marshal.Alloc (free)
import Conllu.Parse (parseConllu)
import Conllu.Type (Doc)

Cpp.context Cpp.cppCtx
Cpp.include "<iostream>"
Cpp.include "<sstream>"
Cpp.include "<udpipe.cpp>"
```

Appendix B. Testing UDPipe

This version keeps reloading the model, which is slow, and also uses UDPipe 1 which is quite bad at languages like Welsh. UDPipe 2 is better, but it

- relies on UDPipe 1 for tokenization
- runs bert-base-multilingual-uncased to extract embeddings
- runs TensorFlow 1 for UD parsing

I'm currently just considering the input as CoNLL-U, wherever it comes from. For English and other well-resourc, UDPipe 1 is probably suff

```
parseOnce :: FilePath → String → String
parseOnce modelPath sentence = unsafePerformIO do
  withCString modelPath \cModel → withCString sentence \cSentence → do
    output ← [Cpp.exp| char* { [&]() -> char* {
    using namespace ufal::udpipe;

    model* m = model::load($(char* cModel));
    if (!m) {
      return strdup("");
    }

    pipeline p(
      m,
      "tokenize",
      pipeline::DEFAULT,
      pipeline::DEFAULT,
      "conllu"
    );

    std::istringstream input($(char* cSentence));
    std::ostringstream output;
    std::string error;

    bool succeeded = p.process(input, output, error);

    delete m;
    return strdup(succeeded ? output.str().c_str() : "");
  }() } |]
    result ← peekCString output
    free output
    pure result

{-# NOINLINE parseOnce #-}
```

Appendix C. Stubs

```
resolve = undefined
dependencyParse = undefined
understand = undefined
explain = undefined
simplify = undefined
interpret = undefined
execute = undefined
view = undefined
webrender = undefined
createFrame = undefined
read = undefined
initialState = undefined
dialectic = undefined
typstLayout = undefined

data Nominal = Nominal
data Noun = Noun
data Sentence = Sentence
data Text = Text
data Semantics = Semantics
data Math = Math
data Input = Input
data Command = Command
data State = State
data DisplayListItem = DisplayListItem
type DisplayList = [DisplayListItem]
data GLContext = GLContext
data Set a = Set
data Fact = Fact
data Knowledge = Knowledge
data Thesis = Thesis
```

References

1. Jake. Fanzines from the 1970s. (2015).
2. Voynov, A., Corbi, A., López-Oliver, P. & Gil Oliva, D. Typst: A Modern Typesetting Engine for Science. (2026).
3. De Marneffe, M.-C., Manning, C. D., Nivre, J. & Zeman, D. Universal dependencies. *Computational linguistics* **47,** 255–308 (2021).